

Unit 2 Review

Date: _____

1. A thin rod of length L carries total charge Q spread uniformly along it. Find the electric field at a point on the rod's axis a distance a from the nearer end.

2. A closed surface encloses point charges $q_1 = +2.0 \text{ nC}$ and $q_2 = -5.0 \text{ nC}$ (and a third charge of $+7.0 \text{ nC}$ sits outside). Find the net electric flux through the surface.
3. An infinitely long line of charge has linear density $\lambda = 2.0 \mu\text{C}/\text{m}$. Find E at a perpendicular distance of 0.050 m .